

ADVANCED

Stage 6

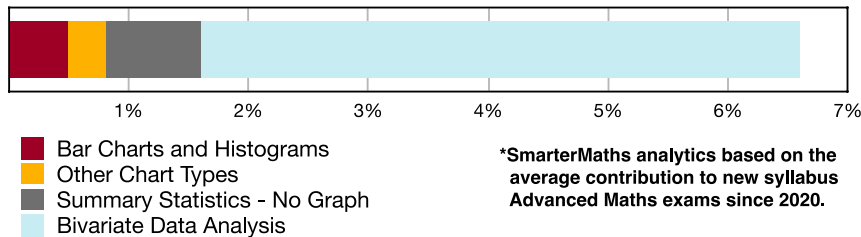
Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Classifying Data (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 10.5 minutes (based on allocation of 1.5 minutes per mark)

S2 Interpretation and Bivariate Data



HISTORICAL CONTRIBUTION

- S2 Interpretation and Bivariate Data* has contributed an average of 6.5% per new syllabus Advanced exam since it began in 2020.
- We have split the area into six sub-topics for analysis purposes which are: 1-*Classifying Data* (0%), 2-*Bar Charts and Histograms* (0.5%), 3-*Other Chart Types* (0.3%), 4-*Summary Statistics - Box Plots* (0%), 5-*Summary Statistics - No Graph* (0.8%) and 6-*Bivariate Data Analysis* (5.0%).
- This analysis looks at *Classifying Data*.

HSC ANALYSIS - What to expect and common pitfalls

- Classifying Data* looks at categorical data (nominal vs ordinal) and numerical data (discrete vs continuous).
- This topic area has not been examined in new syllabus Advanced exams to date although it retains its separate dot point in the syllabus document. It is small area that nonetheless students should cover in case it appears in a multiple choice question (as it has in pre-2020 exams).
- Classifying Data* looks at categorical data (nominal vs ordinal) and numerical data (discrete vs continuous).
- It has been asked in multiple choice questions 3 times since 2016 (in Std2 exams). We note that questions on stratified sampling are beyond the new *Advanced* syllabus and are not included in the database.

Questions

1. Statistics, STD2 S1 2009 HSC 3 MC

The eye colours of a sample of children were recorded.

When analysing this data, which of the following could be found?

- Mean
- Median
- Mode
- Range

2. Statistics, STD2 S1 SM-Bank 1 MC

A survey asked the following question for students born in Australia:

"Which State or Territory were you born in?"

How would the responses be classified?

- Categorical, ordinal
- Categorical, nominal
- Numerical, discrete
- Numerical, continuous

3. Statistics, STD2 S1 2015 HSC 4 MC

On a school report, a student's record of completing homework is graded using the following codes.

C = consistently
U = usually
S = sometimes
R = rarely
N = never

What type of data is this?

- Categorical, ordinal
- Categorical, nominal
- Numerical, continuous
- Numerical, discrete

4. Statistics, STD2 S1 2018 HSC 3 MC

A survey asked the following question.

'How many brothers do you have?'

How would the responses be classified?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, discrete
- D. Numerical, continuous

5. Statistics, STD2 S1 2016 HSC 7 MC

Which set of data is classified as categorical and nominal?

- A. blue, green, yellow
- B. small, medium, large
- C. 5.2 cm, 6 cm, 7.21 cm
- D. 4 people, 5 people, 9 people

6. Statistics, STD2 S1 2011 HSC 25a*

A study on the mobile phone usage of NSW high school students is to be conducted.

Data is to be gathered using a questionnaire.

The questionnaire begins with the three questions shown.

Q1: Do you own a mobile phone?

Yes ☐ No ☐

Q2: Which phone company do you use?

.....

Q3: Do you use pre-paid or a plan?

Pre-paid ☐ Plan ☐

- i. Classify the type of data that will be collected in Q2 of the questionnaire. **(1 mark)**
- ii. Write a suitable question for this questionnaire that would provide discrete ordinal data. **(1 mark)**

Worked Solutions

1. Statistics, STD2 S1 2009 HSC 3 MC

Eye colour is categorical data

∴ Only the mode can be found

⇒ C

2. Statistics, STD2 S1 SM-Bank 1 MC

The data is categorical (not numerical) since the name of a State is required.

This data cannot be ordered.

⇒ B

3. Statistics, STD2 S1 2015 HSC 4 MC

The data has been grouped into categories and because each category can be ranked, it is ordinal.

⇒ A

4. Statistics, STD2 S1 2018 HSC 3 MC

The number of brothers a person has is an exact whole number.

∴ Classification is numerical, discrete.

⇒ C

5. Statistics, STD2 S1 2016 HSC 7 MC

Categorical and nominal data is qualitative and not ordered.

⇒ A

♦♦ Mean mark 26%.

6. Statistics, STD2 S1 2011 HSC 25a*

- i. Categorical, nominal
- ii. How many outgoing calls do you make per day?
(Ensure it can be answered with a numerical score.)